# Worksheet 3.4 — Radian Measure, Arcs and Sectors

Mathematics: Analysis & Approaches SL (IB) · Unit 3: Geometry and Trigonometry

### The Big Ideas

Radians make arc length and sector area formulas clean: $s = r\theta$, $A = \frac{1}{2}r^2\theta$.

- $180° = \pi$ rad.
- Segment area $= \frac{1}{2}r^2(\theta - \sin\theta)$.

## Worked Examples

### Example 1 — Convert degrees to radians

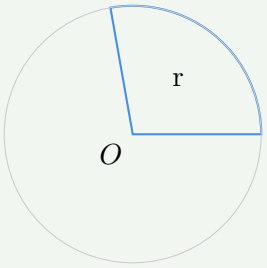
Convert 315° to radians as an exact multiple of $\pi$.

***Solution.*** $315 \times \dfrac{\pi}{180} = \dfrac{7\pi}{4}$.

**Conclusion:** $\dfrac{7\pi}{4}$.

### Example 2 — Solve for the angle

An arc of length 15 cm is cut from a circle of radius 6 cm. Find $\theta$ in radians.




***Solution.*** $15 = 6\theta \Rightarrow \theta = 2.5$.

**Conclusion:** $\theta = 2.5$ rad.

### Example 3 — Solve for the radius

A sector has area 45 cm$^2$ and angle $\dfrac{3\pi}{4}$. Find $r$.

***Solution.*** $45 = \frac{1}{2}r^2\left(\frac{3\pi}{4}\right) \Rightarrow r^2 = \dfrac{120}{\pi}$.

**Conclusion:** $r \approx 6.18$ cm.

### Example 4 — Segment area

Find the segment area for a circle of radius 8 cm with central angle $\dfrac{\pi}{2}$.

***Solution.*** Sector: $\frac{1}{2}(64)(\pi/2) = 16\pi$. Triangle: $\frac{1}{2}(64)\sin(\pi/2) = 32$.

**Conclusion:** $16\pi - 32 \approx 18.27$ cm$^2$.

### Example 5 — Convert a non-standard radian value

Convert 2 radians to degrees (nearest 0.1ř).

***Solution.*** $2 \times \dfrac{180}{\pi}$.

**Conclusion:** $\approx 114.6$ř.

### Example 6 — Application — clock hand arc

A minute hand of length 8 cm sweeps from 12 to 4 (one third of a full circle). Find the arc length traced by its tip.

***Solution.*** One third of a circle is $\dfrac{2\pi}{3}$ rad. $s = 8\left(\dfrac{2\pi}{3}\right)$.

**Conclusion:** $s = \dfrac{16\pi}{3} \approx 16.76$ cm.

### Example 7 — Application — pizza slice area

A pizza of radius 15 cm is cut into a slice with central angle 50ř. Find the slice's area.

***Solution.*** Convert: $\theta = 50 \times \dfrac{\pi}{180} = \dfrac{5\pi}{18} \approx 0.8727$ rad. Area $= \frac{1}{2}(225)(0.8727)$.

**Conclusion:** area $\approx 98.18$ cm$^2$.

### Example 8 — Application — circular window segment

A circular window of radius 40 cm has a chord cutting off a segment with central angle $\dfrac{2\pi}{5}$. Find the segment's area (nearest cm$^2$).

***Solution.*** Sector: $\frac{1}{2}(1600)\left(\frac{2\pi}{5}\right) = 320\pi \approx 1005.3$. Triangle: $\frac{1}{2}(1600)\sin\left(\frac{2\pi}{5}\right) \approx 760.9$.

**Conclusion:** segment area $\approx 244$ cm$^2$.

### Example 9 — Sector area from arc length and radius

A sector's arc length is 20 cm and its radius is 8 cm. Find the sector's area.

***Solution.*** $\theta = \dfrac{20}{8} = 2.5$ rad. Area $= \frac{1}{2}(64)(2.5)$.

**Conclusion:** area $= 80$ cm$^2$.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Convert 250ř to radians (round to 3 d.p.).

### Question 2

Convert $\frac{5\pi}{9}$ radians to degrees.

### Question 3

Find the arc length for angle 1.4 rad in a circle of radius 9 cm.

### Question 4

A sector has area 60 $\text{cm}^2$ and radius 10 cm; find its angle in radians.

### Question 5 — Challenge

Find the segment area for a circle of radius 12 cm with central angle $\frac{2\pi}{3}$.

## Question 6 — Challenge

A sector's perimeter is 30 cm and its area is 54 $\text{cm}^2$; set up and solve for its radius $r$.

## Question 7

A running track is a rectangle 90 m by 24 m, capped by two semicircular ends of radius 12 m; find the total perimeter.

## Question 8

Find the area of a sector with radius 5 cm and angle 2.2 rad.

## Question 9

Convert 1.5 radians to degrees (nearest 0.1ř).

### Question 10

A clock's hour hand (length 6 cm) sweeps from 12 to 3 (one quarter of a circle); find the arc length traced.

### Question 11 — Challenge

A pizza of radius 20 cm is cut into a slice with central angle 40ř; find the slice's area (nearest $\text{cm}^2$).

### Question 12 — Challenge

A circular window of radius 30 cm has a segment with central angle $\frac{\pi}{3}$; find the segment's area (nearest $\text{cm}^2$).

### Question 13 — Challenge

A sector's arc length is 18 cm, radius 6 cm; find its area.

# Question Recap & Answer Key

## Question Recap

1. Convert 250ř to radians (round to 3 d.p.).
2. Convert $\frac{5\pi}{9}$ radians to degrees.
3. Find the arc length for angle 1.4 rad in a circle of radius 9 cm.
4. A sector has area 60 cm$^2$ and radius 10 cm; find its angle in radians.
5. Find the segment area for a circle of radius 12 cm with central angle $\frac{2\pi}{3}$.
6. A sector's perimeter is 30 cm and its area is 54 cm$^2$; set up and solve for its radius $r$.
7. A running track is a rectangle 90 m by 24 m, capped by two semicircular ends of radius 12 m; find the total perimeter.
8. Find the area of a sector with radius 5 cm and angle 2.2 rad.
9. Convert 1.5 radians to degrees (nearest 0.1ř).
10. A clock's hour hand (length 6 cm) sweeps from 12 to 3 (one quarter of a circle); find the arc length traced.
11. A pizza of radius 20 cm is cut into a slice with central angle 40ř; find the slice's area (nearest cm$^2$).
12. A circular window of radius 30 cm has a segment with central angle $\frac{\pi}{3}$; find the segment's area (nearest cm$^2$).
13. A sector's arc length is 18 cm, radius 6 cm; find its area.

## ✓ Answer Key

1. $250 \times \pi/180 \approx 4.363$
2. $5\pi/9 \times 180/\pi = 100$ř
3. $s = 9(1.4) = 12.6$ cm
4. $60 = \frac{1}{2}(100)\theta \Rightarrow \theta = 1.2$ rad
5. sector $= \frac{1}{2}(144)(2\pi/3) = 48\pi$; triangle $= \frac{1}{2}(144)\sin(2\pi/3) = 72 \cdot \frac{\sqrt{3}}{2} \approx 62.35$; segment $\approx 150.80 - 62.35 \approx 88.45$ cm$^2$
6. $2r + r\theta = 30$, $\frac{1}{2}r^2\theta = 54 \Rightarrow r\theta = 108/r$; $2r + 108/r = 30 \Rightarrow 2r^2 - 30r + 108 = 0 \Rightarrow r^2 - 15r + 54 = 0 \Rightarrow (r-6)(r-9) = 0$: $r = 6$ or $r = 9$
7. straight sides $2(90) = 180$; two semicircles $= 2(12\pi) = 24\pi$; total $\approx 180 + 75.4 = 255.4$ m
8. $\frac{1}{2}(25)(2.2) = 27.5$ cm$^2$
9. $1.5 \times 180/\pi \approx 85.9$ř
10. quarter circle $= \pi/2$ rad; $s = 6(\pi/2) = 3\pi \approx 9.42$ cm
11. $\theta = 40\pi/180 = 2\pi/9 \approx 0.6981$; area $= \frac{1}{2}(400)(0.6981) \approx 140$ cm$^2$
12. sector $= \frac{1}{2}(900)(\pi/3) = 150\pi \approx 471.24$; triangle $= \frac{1}{2}(900)\sin(\pi/3) \approx 389.71$; segment $\approx 82$ cm$^2$
13. $\theta = 18/6 = 3$ rad; area $= \frac{1}{2}(36)(3) = 54$ cm$^2$